Executive Summary

This module commands a sequence of constant-rate rotations about user-specified body axes. The module supports `kNumRotations` (4) consecutive rotation maneuvers. Each rotation is defined by a rotation axis, a commanded body rate magnitude, and a duration. The axes are user defined and any combination of the three principal axes is allowed, including repeated rotations about the same axis.

The module reads the spacecraft's current angular velocity and computes a guidance message containing the reference rate `omega_RN_B`, the body-rate error `omega_BR_B`, and zero reference angular acceleration. The output of this module should be paired with a rate control module such as `rateControl`.

Message Connection Descriptions

The following table lists all the module input and output messages.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
attNavInMsg	`NavAttMsgF32Payload`	Input message containing current spacecraft rates in body-frame coordinates.
attGuidOutMsg	`AttGuidMsgF32Payload`	Output guidance message containing reference and error rates.

Algorithm Inputs and Outputs

Algorithm inputs:

Variable Name	Description
callTime	Current simulation time in nanoseconds, used to determine which rotation is active.
omega_BN_B	Measured spacecraft angular velocity in the body frame $\boldsymbol{\omega}_{B/N}$.

Algorithm outputs:

Variable Name	Description
omega_RN_B	Reference angular velocity in the body frame $\boldsymbol{\omega}_{R/N}$. Held at the last rotation's commanded rate after the sequence completes.
omega_BR_B	Body-rate error in the body frame $\boldsymbol{\omega}_{B/R} = \boldsymbol{\omega}_{B/N} - \boldsymbol{\omega}_{R/N}$.

Detailed Module Description

This module is intended to be used when the spacecraft might not have a valid Sun direction estimate. It commands a sequence of rotations to sweep the body-fixed Sun sensors through different orientations until the Sun is found. The search pattern consists of exactly `kNumRotations` (4) consecutive rotation maneuvers. Each maneuver rotates the spacecraft at a constant rate about a user-specified body axis for a user-specified duration. The four rotations execute sequentially: rotation 1 runs for its full duration, then rotation 2 begins, and so on.

For each active rotation, the module outputs a constant reference angular velocity `omega_RN_B` along the corresponding body axis. The body-rate error is computed as:

$$\boldsymbol{\omega}_{B/R} = \boldsymbol{\omega}_{B/N} - \boldsymbol{\omega}_{R/N}$$

where $\boldsymbol{\omega}_{B/N}$ is the measured body rate from the navigation message and $\boldsymbol{\omega}_{R/N}$ is the commanded reference rate. The body-rate error is computed here because the `attitudeTrackingError` module is not used in Safe mode.

After all four rotations have completed, the algorithm continues to command the last rotation's angular velocity indefinitely. This keeps the spacecraft rotating if the Sun has not been acquired by the end of the scripted sequence; the final rotation's rate and axis therefore double as the indefinite-hold configuration.

Each rotation's `rotationDuration` must be finite and strictly greater than zero. Zero, negative, NaN, and infinite values are rejected at configuration time. Internally the algorithm accumulates rotation end-times in 64-bit integer nanoseconds, so cumulative durations are exact at the precision of the input.

User Guide

The required module configuration is:

```
attGuid = sunSearchF32.SunSearch()

rotationProp = sunSearchF32.RotationProperties()
```

```
rotationProp.rotationDuration = 5.0
rotationProp.rotationRate = 0.1
rotationProp.rotationAxis = sunSearchF32.RotationAxis_b1Hat_B
attGuid.setRotation(0, rotationProp)

rotationProp = sunSearchF32.RotationProperties()
rotationProp.rotationDuration = 10.0
rotationProp.rotationRate = 0.2
rotationProp.rotationAxis = sunSearchF32.RotationAxis_b2Hat_B
attGuid.setRotation(1, rotationProp)

rotationProp = sunSearchF32.RotationProperties()
rotationProp.rotationDuration = 7.0
rotationProp.rotationRate = 0.15
rotationProp.rotationAxis = sunSearchF32.RotationAxis_b3Hat_B
attGuid.setRotation(2, rotationProp)

rotationProp = sunSearchF32.RotationProperties()
rotationProp.rotationDuration = 3.0
rotationProp.rotationRate = 0.05
rotationProp.rotationAxis = sunSearchF32.RotationAxis_b1Hat_B
attGuid.setRotation(3, rotationProp)
```

To retrieve or modify a rotation after it has been set:

```
rotation = attGuid.getRotation(1)
rotation.rotationRate = 0.3
attGuid.setRotation(1, rotation)
```